

Application of 3-phase network :

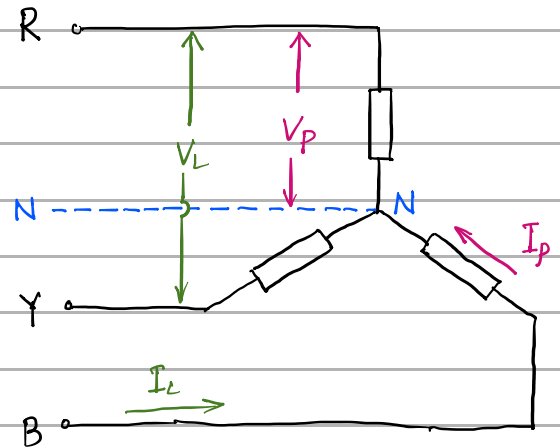
- High power motors
- Power transformers
- Distribution transformers
- AC Power transmission
- Other industrial applications.

Types of 3-phase connection :

(I) star connection :

$$\left. \begin{array}{l} V_L = \text{line voltage} \\ V_P = \text{phase voltage} \end{array} \right\} V_P = \frac{V_L}{\sqrt{3}}$$

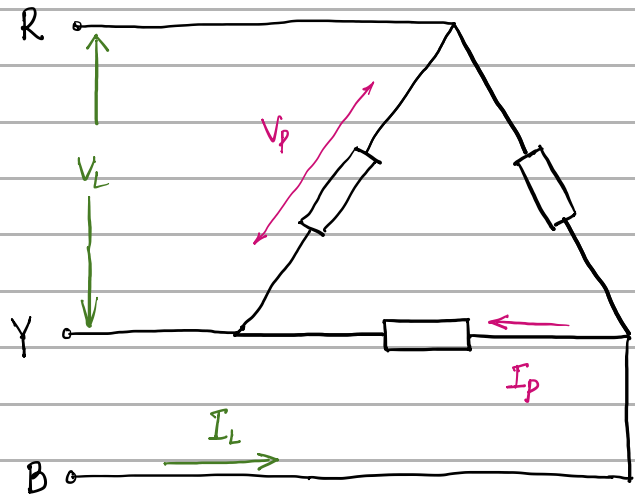
$$\left. \begin{array}{l} I_L = \text{line current} \\ I_P = \text{phase current} \end{array} \right\} I_P = I_L$$



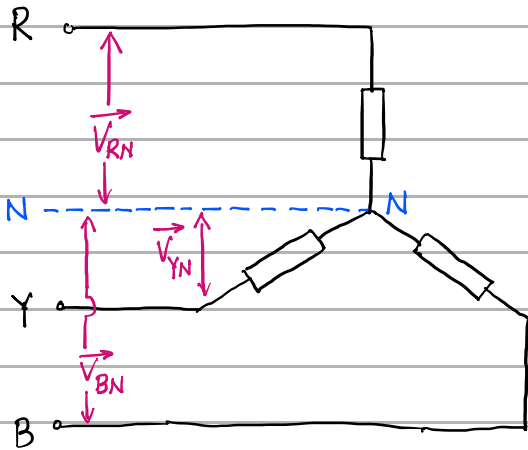
(II) Delta connection :

$$\left. \begin{array}{l} V_L = \text{line voltage} \\ V_P = \text{phase voltage} \end{array} \right\} V_P = V_L$$

$$\left. \begin{array}{l} I_L = \text{line current} \\ I_P = \text{phase current} \end{array} \right\} I_P = \frac{I_L}{\sqrt{3}}$$



Representation of voltage in 3-phase circuits:



The magnitude of voltages in all three phases are equal but they are 120° phase displaced.

If V_{RN} is considered as reference voltage and RYB is the phase sequence,

$$V_{RN} = V_m \sin \omega t \quad (\text{ref.})$$

$$V_{YN} = V_m \sin(\omega t - 120^\circ)$$

$$V_{BN} = V_m \sin(\omega t - 240^\circ)$$

$$\text{or } V_m \sin(\omega t + 120^\circ)$$

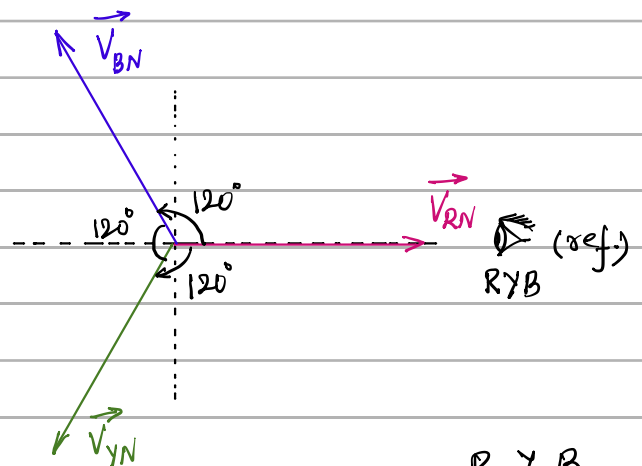
phase
voltages

In terms of phasors,

$$\vec{V}_{RN} = V_{rms} \angle 0^\circ$$

$$\vec{V}_{YN} = V_{rms} \angle -120^\circ$$

$$\vec{V}_{BN} = V_{rms} \angle -240^\circ$$



R Y B

Line voltage is evaluated as,

$$\vec{V}_{RY} = \vec{V}_{RN} - \vec{V}_{YN} = V_{rms} \angle 0^\circ - V_{rms} \angle -120^\circ$$

$$= V_{rms} + \frac{1}{2} V_{rms} + j \frac{\sqrt{3}}{2} V_{rms}$$

$$= \sqrt{3} V_{rms} \angle 30^\circ$$

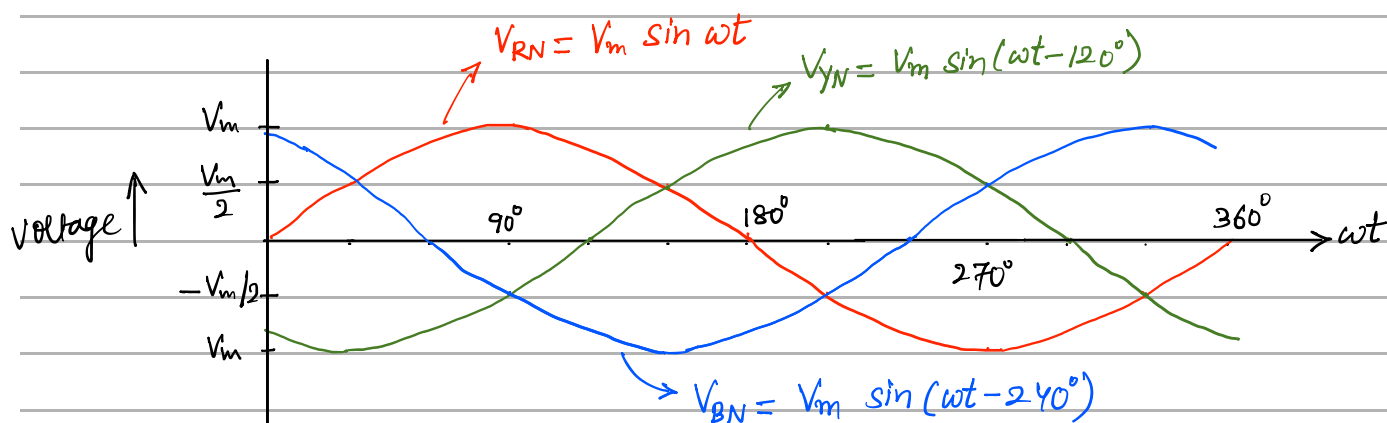
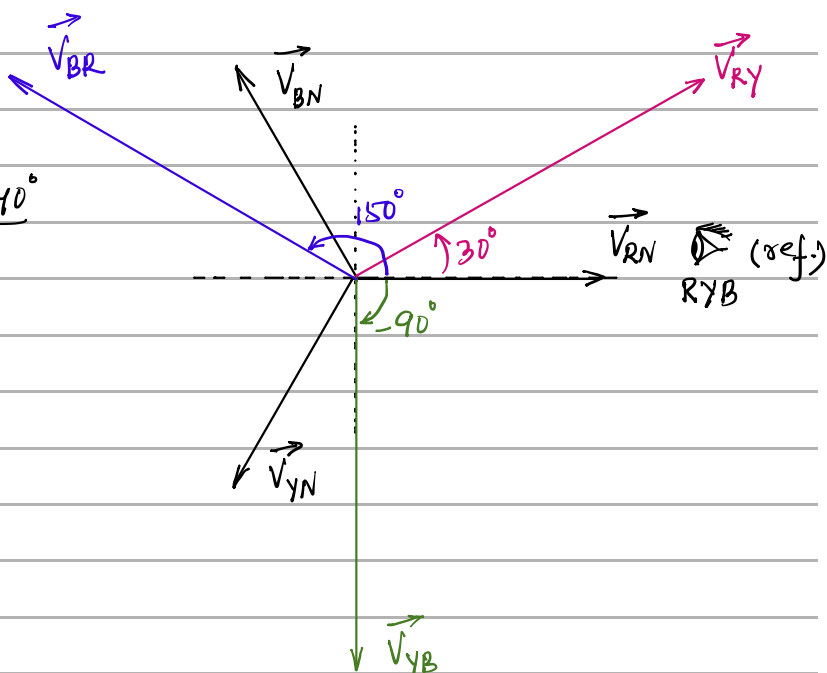
$$\left. \begin{array}{l} \text{Note: } |\vec{V}_L| = |\vec{V}_{RY}| = \sqrt{3} V_{rms} \\ |\vec{V}_P| = |\vec{V}_{RN}| = V_{rms} \end{array} \right\} V_P = \frac{V_L}{\sqrt{3}}$$

Similarly,

$$\begin{aligned}\vec{V}_{YB} &= \vec{V}_{YN} - \vec{V}_{BN} \\ &= V_{rms} \angle -120^\circ - V_{rms} \angle -240^\circ \\ &= \sqrt{3} V_{rms} \angle -90^\circ\end{aligned}$$

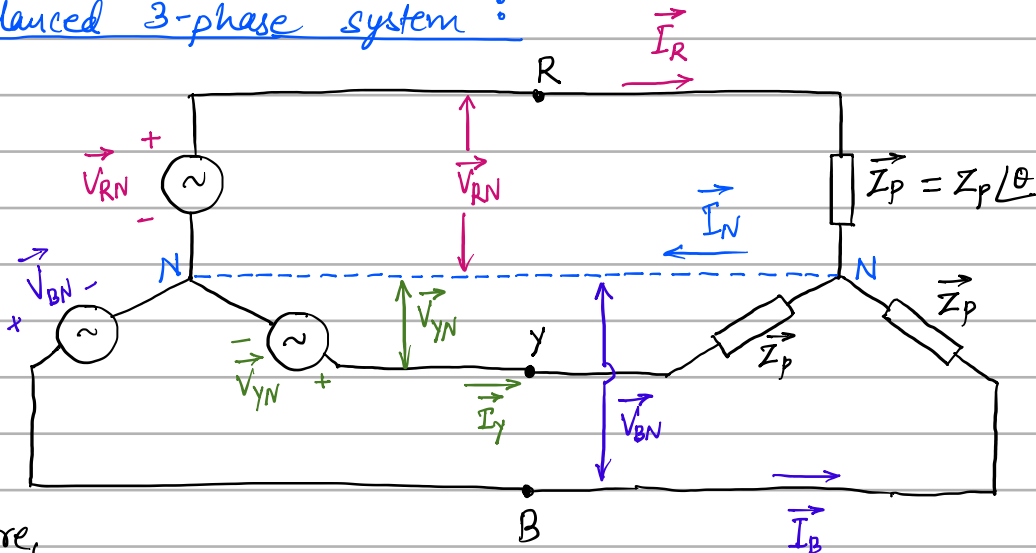
and,

$$\begin{aligned}\vec{V}_{BR} &= \vec{V}_{BN} - \vec{V}_{RN} \\ &= V_{rms} \angle -240^\circ - V_{rms} \angle 0^\circ \\ &= \sqrt{3} V_{rms} \angle 150^\circ\end{aligned}$$



Balanced 3-phase system :

Imp



Here,

$$\vec{V}_{RN} = V_{rms} \angle 0^\circ$$

$$\vec{V}_{YN} = V_{rms} \angle -120^\circ$$

$$\vec{V}_{BN} = V_{rms} \angle -240^\circ$$

$$\text{So, } \vec{I}_R = \frac{\vec{V}_{RN}}{\vec{Z}_p} = \frac{V_{rms} \angle 0^\circ}{Z_p \angle \theta} = \frac{V_{rms}}{Z_p} \angle -\theta = I_{rms} \angle -\theta$$

$$\vec{I}_Y = \frac{\vec{V}_{YN}}{\vec{Z}_p} = \frac{V_{rms} \angle -120^\circ}{Z_p \angle \theta} = \frac{V_{rms}}{Z_p} \angle -120^\circ - \theta = I_{rms} \angle (-\theta) - 120^\circ$$

$$\vec{I}_B = \frac{\vec{V}_{BN}}{\vec{Z}_p} = \frac{V_{rms} \angle -240^\circ}{Z_p \angle \theta} = \frac{V_{rms}}{Z_p} \angle -240^\circ - \theta = I_{rms} \angle (-\theta) - 240^\circ$$

Therefore, in a balanced 3- ϕ circuit, the current in each phase are equal in magnitude but they are 120° phase displaced.

Now, current through the neutral,

$$\vec{I}_N = \vec{I}_R + \vec{I}_Y + \vec{I}_B$$

$$= I_{rms} \angle -\theta + I_{rms} \angle (-\theta) - 120^\circ + I_{rms} \angle (-\theta) - 240^\circ = 0$$

The neutral carries no current if the source and load both are balanced.

Three-phase power :

Imp Total power consumed by a 3-phase system (both star & delta)

= sum of power associated with each phase

$$\text{SO, } \vec{S} = \vec{V}_{RN} \vec{I}_R^* + \vec{V}_{YN} \vec{I}_Y^* + \vec{V}_{BN} \vec{I}_B^* \quad (\text{refer to diagram on Page-4})$$

$$= V_{rms} \angle 0^\circ \cdot I_{rms} \angle \theta + V_{rms} \angle -120^\circ \cdot I_{rms} \angle +\theta + 120^\circ + V_{rms} \angle -240^\circ \cdot I_{rms} \angle +\theta + 240^\circ$$

$$= V_{rms} I_{rms} \angle \theta + V_{rms} \cdot I_{rms} \angle \theta + V_{rms} \cdot I_{rms} \angle \theta$$

$$= 3 V_{rms} I_{rms} \angle \theta$$

$$\begin{cases} Z_p \angle \theta \rightarrow \text{Inductive load} \\ Z_p \angle -\theta \rightarrow \text{Capacitive load} \end{cases}$$

$$= 3 V_p I_p \angle \theta$$

$$= 3 \vec{V}_p \vec{I}_p^*$$

$$\text{SO, } \boxed{\vec{S} = 3 \vec{V}_p \vec{I}_p^*} = 3 V_p I_p \cos \phi + j 3 V_p I_p \sin \phi$$

$$\text{And, active power, } P = 3 V_p I_p \cos \theta = \sqrt{3} V_L I_L \cos \theta \quad \left[V_p = V_L / \sqrt{3} \right] \quad (\text{star})$$

$$\text{Reactive power, } Q = 3 V_p I_p \sin \theta = \sqrt{3} V_L I_L \sin \theta$$

where, θ = angle between current ($\vec{I}_p$) & voltage phasor ($\vec{V}_p$)

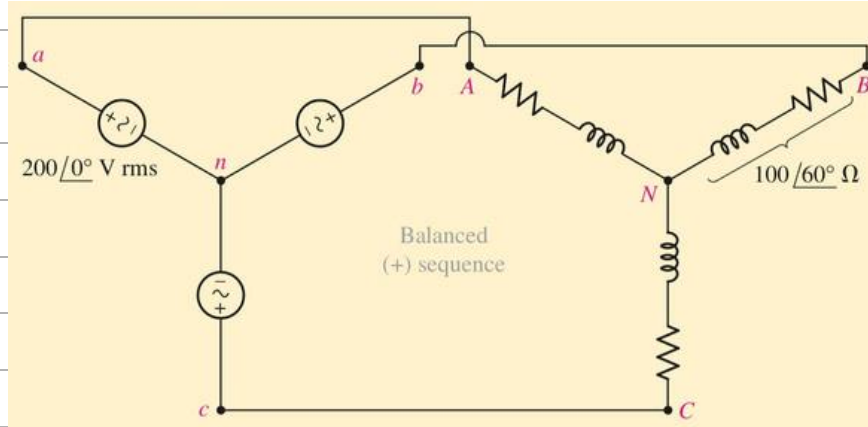
— The above expressions are valid for both star and delta connections.

Q Find the following:

(a) Phase voltages and line voltages

(b) Phase current and line currents.

(c) Calculate the power dissipated in load.



Solⁿ:

Given, phase voltage, $\vec{V}_{an} = 200 \angle 0^\circ \text{ V}$

$$\vec{V}_{bn} = 200 \angle -120^\circ \text{ V}$$

$$\vec{V}_{cn} = 200 \angle -240^\circ \text{ V}$$

phase voltages

So, line voltages,

$$\begin{aligned} \vec{V}_{ab} &= \vec{V}_{an} - \vec{V}_{bn} \\ &= 200 \angle 0^\circ - 200 \angle -120^\circ \text{ V} \\ &= 200\sqrt{3} \angle 30^\circ \text{ V} \end{aligned}$$

$$\begin{aligned} \vec{V}_{bc} &= \vec{V}_{bn} - \vec{V}_{cn} \\ &= 200 \angle -120^\circ - 200 \angle -240^\circ \text{ V} \\ &= 200\sqrt{3} \angle -90^\circ \text{ V} \end{aligned}$$

$$\begin{aligned} \vec{V}_{ca} &= \vec{V}_{cn} - \vec{V}_{an} \\ &= 200 \angle -240^\circ - 200 \angle 0^\circ \text{ V} \\ &= 200\sqrt{3} \angle 150^\circ \text{ V} \end{aligned}$$

Line voltages

Here, impedance per phase = $\vec{Z} = 100 \angle 60^\circ \Omega$

So, phase (=line) currents are calculated as,

$$\vec{I}_{aA} = \frac{\vec{V}_{an}}{\vec{Z}} = \frac{200 \angle 0^\circ}{100 \angle 60^\circ} = 2 \angle -60^\circ \text{ A}$$

$$\vec{I}_{bB} = \frac{\vec{V}_{bn}}{\vec{Z}} = \frac{200 \angle -120^\circ}{100 \angle 60^\circ} = 2 \angle -180^\circ \text{ A}$$

$$\vec{I}_{cC} = \frac{\vec{V}_{cn}}{\vec{Z}} = \frac{200 \angle -240^\circ}{100 \angle 60^\circ} = 2 \angle -300^\circ \text{ A}$$

Line / phase currents

Power dissipated per phase = $V_{\text{phase}} \cdot I_{\text{phase}} \cos \phi$ ($\phi = \text{angle between } \vec{V}_{\text{phase}} \text{ \& } \vec{I}_{\text{phase}}$)

$$= 200 \times 2 \times \cos(0^\circ + 60^\circ)$$

or, $-120^\circ + 180^\circ$
or, $-240^\circ + 300^\circ$

$$= 400 \cos 60^\circ$$

$$= 400 \times \frac{1}{2}$$

$$= 200 \text{ watt} = P_{\text{phase}}$$

So, total (active) power in all three phases = $3 \times 200 = 600 \text{ watt}$
 $= P_{\text{total}}$

(OR),

Total power dissipated in all three phases,

$$P_{\text{total}} = \sqrt{3} V_{\text{line}} I_{\text{line}} \cos \phi$$

$$= \sqrt{3} \times 200 \sqrt{3} \times 2 \times \cos(60^\circ)$$

$$= 600 \text{ watt}$$

(OR),

$$\vec{S} = 3 \vec{V}_{\text{phase}} \cdot \vec{I}_{\text{phase}}^*$$

$$= 3 \times 200 \angle 0^\circ \times 2 \angle +60^\circ$$

$$= 1200 \angle 60^\circ = 600 + j1039.23$$

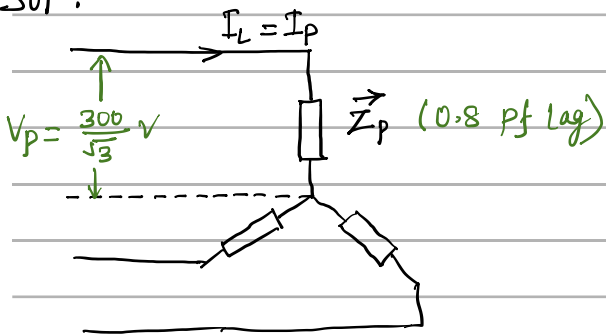
Apparent
Power (VA)

Active
power (W)

Reactive
power (VAR)

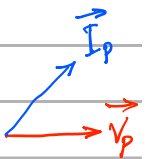
Q A balanced 3-phase system with a line voltage of 300V is supplying a balanced Y-connected load with 1200W at leading PF of 0.8. Find the line current and per-phase load impedance.

Solⁿ:



$$\text{Phase voltage, } \vec{V}_p = \frac{300}{\sqrt{3}} \angle 0^\circ \text{ V}$$

$$\vec{I}_p = I_p \angle +\cos^{-1}(0.8) \text{ A}$$



Also,

$$1200 \text{ W} = 3 V_p I_p \cos \phi$$

$$\Rightarrow 1200 = 3 \times \frac{300}{\sqrt{3}} \cdot I_p \times 0.8$$

$$\Rightarrow I_p = 2.887 \text{ A}$$

$$\therefore \vec{I}_p = 2.887 \angle \cos^{-1}(0.8) \text{ A} = \vec{I}_L = 2.887 \angle 36.89^\circ \text{ A}$$

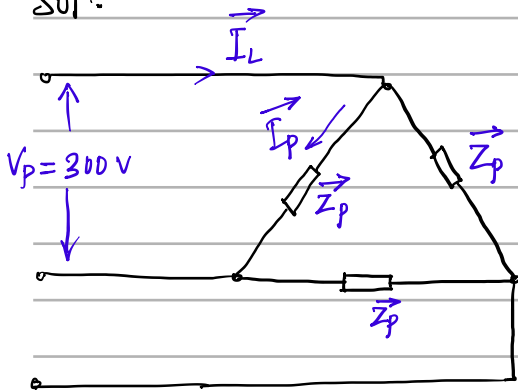
$$\vec{Z}_p = \frac{\vec{V}_p}{\vec{I}_p} = \frac{300/\sqrt{3} \angle 0^\circ}{2.887 \angle 36.89^\circ} \Omega = 36.869^\circ$$

$$\Rightarrow \vec{Z}_p = 60 \angle -36.89^\circ \Omega \approx 48 - j36 \Omega$$

Imp

Q A balanced 3-phase system with a line voltage of 300V is supplying a balanced Δ -connected load with 1200W at lagging PF of 0.8. Find the line current and per-phase load impedance.

Soln:



$$\text{Phase voltage, } \vec{V}_p = 300 \angle 0^\circ \text{ V}$$

$$\vec{I}_p = I_p \angle -\cos^{-1}(0.8) \text{ A}$$

And,

$$1200 = 3 V_p I_p \cos \phi$$

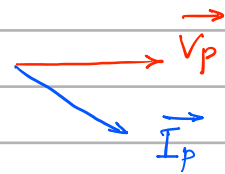
$$\Rightarrow 1200 = 3 \times 300 \times I_p \times 0.8$$

$$\Rightarrow I_p = 1.67 \text{ A}$$

$$\text{So, } I_L = 1.67 \sqrt{3}$$

$$\Rightarrow I_L = 2.89 \text{ A}$$

$$\vec{Z}_p = \frac{\vec{V}_p}{\vec{I}_p} = \frac{300 \angle 0^\circ}{1.67 \angle -\cos^{-1}(0.8)}$$



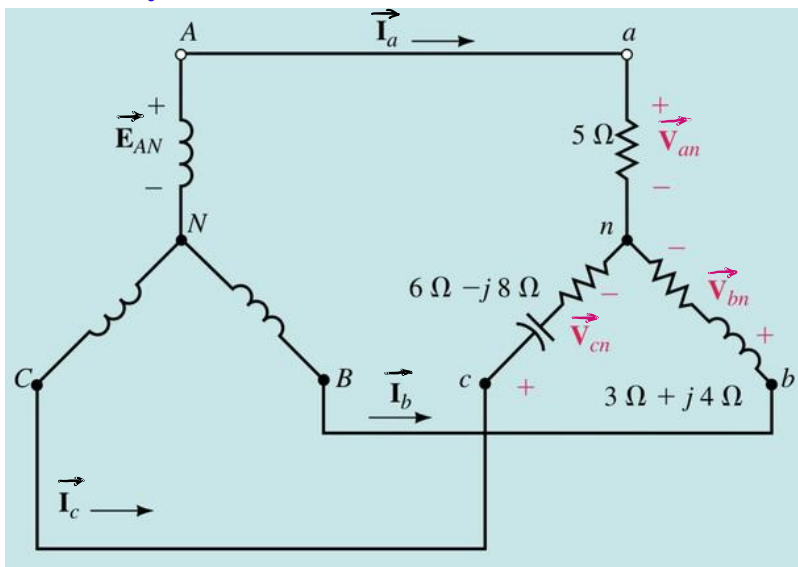
$$\Rightarrow \vec{Z}_p = 180 \angle 36.89^\circ \Omega$$

Unbalanced 3-phase loads:

If the loads in at least one of the phase is different, the 3-phase circuit is called unbalanced.

For unbalanced loads, none of the balanced circuit relationships are applicable. Therefore, each problem must be treated as a 3-phase problem which are analysed using KCL and KVL.

Q The generator is balanced with line-to-line voltage of 208 V. Taking $\vec{E}_{AB}$ as reference, determine line currents and load voltages.



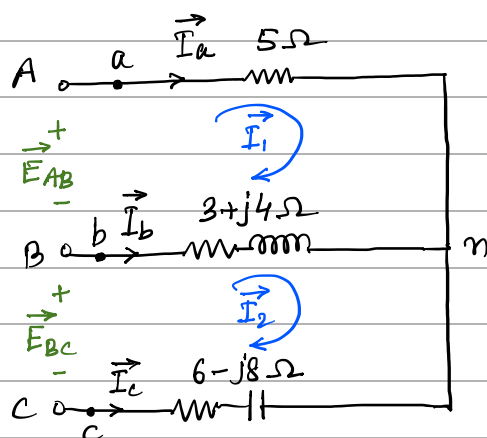
Solⁿ:

$$\text{So, } \vec{E}_{AB} = 208 \angle 0^\circ \text{ V}$$

$$\vec{E}_{BC} = 208 \angle -120^\circ \text{ V}$$

$$\vec{E}_{CA} = 208 \angle -240^\circ \text{ V}$$

We now redraw the circuit and show the mesh currents $\vec{I}_1$ and $\vec{I}_2$:



KVL in Mesh 1:

$$\vec{E}_{AB} = \vec{I}_1 (5) + (\vec{I}_1 - \vec{I}_2) (3 + j4)$$

$$\Rightarrow (5 + 3 + j4) \vec{I}_1 - (3 + j4) \vec{I}_2 = 208 \angle 0^\circ$$

$$\Rightarrow (8 + j4) \vec{I}_1 - (3 + j4) \vec{I}_2 = 208 \angle 0^\circ \quad \text{--- (i)}$$

KVL in Mesh 2:

$$\vec{E}_{BC} = (\vec{I}_2 - \vec{I}_1) (3 + j4) + (6 - j8) \vec{I}_2$$

$$\Rightarrow -(3 + j4) \vec{I}_1 + (3 + j4 + 6 - j8) \vec{I}_2 = 208 \angle -120^\circ$$

$$\Rightarrow -(3 + j4) \vec{I}_1 + (9 - j4) \vec{I}_2 = 208 \angle -120^\circ \quad \text{--- (ii)}$$

Solving (i) & (ii) using calculator, we get

$$\vec{I}_1 = 29.85 \angle -26.21^\circ \text{ A}$$

$$\vec{I}_2 = 11.81 \angle -51.55^\circ \text{ A}$$

So, line currents are calculated as:

$$\vec{I}_a = \vec{I}_1 = 29.85 \angle -26.21^\circ \text{ A}$$

$$\vec{I}_b = \vec{I}_2 - \vec{I}_1 = 11.81 \angle -51.55^\circ - 29.85 \angle -26.21^\circ = 19.83 \angle 168.56^\circ \text{ A}$$

$$\vec{I}_c = -\vec{I}_2 = -11.81 \angle -51.55^\circ = 11.81 \angle 128.45^\circ \text{ A}$$

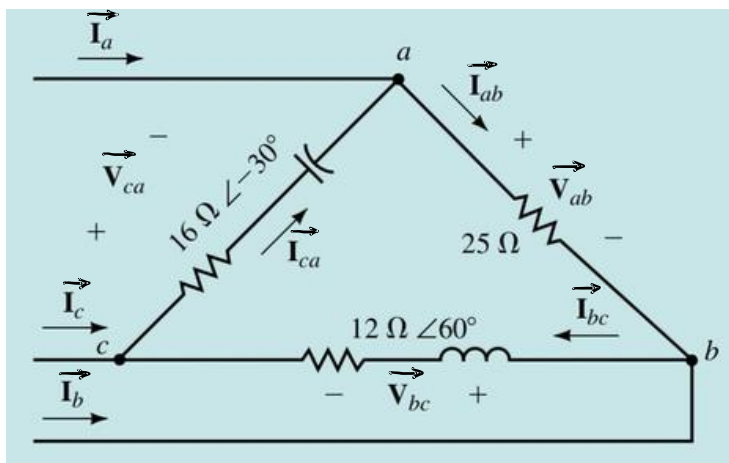
And, load voltages are calculated as:

$$\vec{V}_{an} = 5\vec{I}_a = 5 \times 29.85 \angle -26.21^\circ = 149.25 \angle -26.21^\circ \text{ V}$$

$$\vec{V}_{bn} = (3+j4)\vec{I}_b = (3+j4) \times 19.83 \angle 168.56^\circ = 99.15 \angle -138.31^\circ \text{ V}$$

$$\vec{V}_{cn} = (6-j8)\vec{I}_c = (6-j8) \times 11.81 \angle 128.45^\circ = 118.1 \angle 75.32^\circ \text{ V}$$

Q An unbalanced delta connected load is powered by 3-phase, 240 V, 50 Hz supply. Taking $\vec{V}_{ab}$ as reference, find



- (i) phase currents and draw their phasor diagram
- (ii) line currents
- (iii) Total power to the load.

Solⁿ: $\vec{V}_{ab} = 240 \angle 0^\circ$ volt

$$\vec{V}_{bc} = 240 \angle -120^\circ \text{ volt}$$

$$\vec{V}_{ca} = 240 \angle -240^\circ \text{ volt}$$

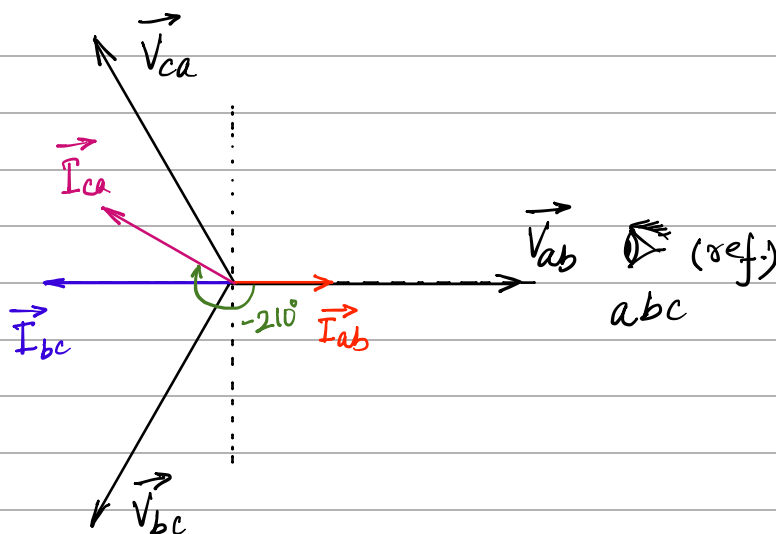
(i) Calculation of phase currents:

$$\vec{I}_{ab} = \frac{\vec{V}_{ab}}{25} = \frac{240 \angle 0^\circ}{25} = 9.6 \angle 0^\circ \text{ A}$$

$$\vec{I}_{bc} = \frac{\vec{V}_{bc}}{12 \angle 60^\circ} = \frac{240 \angle -120^\circ}{12 \angle 60^\circ} = 20 \angle -180^\circ \text{ A} \quad \text{or } 20 \angle 180^\circ \text{ A}$$

$$\vec{I}_{ca} = \frac{\vec{V}_{ca}}{16 \angle -30^\circ} = \frac{240 \angle -240^\circ}{16 \angle -30^\circ} = 15 \angle -210^\circ \text{ A} \quad \text{or } 15 \angle 150^\circ \text{ A}$$

phasor diagram:



(ii) Line currents :

KCL at node a:

$$\vec{I}_a + \vec{I}_{ca} = \vec{I}_{ab}$$

$$\Rightarrow \vec{I}_a = \vec{I}_{ab} - \vec{I}_{ca} = 9.6 \angle 0^\circ - 15 \angle -210^\circ = 23.80 \angle -18.37^\circ \text{ A}$$

KCL at node b:

$$\vec{I}_b = \vec{I}_{bc} - \vec{I}_{ab} = 20 \angle -180^\circ - 9.6 \angle 0^\circ = 29.6 \angle 180^\circ \text{ A}$$

KCL at node c:

$$\vec{I}_c = \vec{I}_{ca} - \vec{I}_{bc} = 15 \angle 150^\circ - 20 \angle -180^\circ = 10.27 \angle 46.94^\circ \text{ A}$$

(iii) Total power to the load:

$$\vec{S}_{ab} = \vec{V}_{ab} \vec{I}_{ab}^* = 240 \angle 0^\circ \times 9.6 \angle 0^\circ = 2304 + j0 \text{ VA}$$

$$\vec{S}_{bc} = \vec{V}_{bc} \vec{I}_{bc}^* = 240 \angle -120^\circ \times 20 \angle 180^\circ = 2400 + j4156.92 \text{ VA}$$

$$\vec{S}_{ca} = \vec{V}_{ca} \vec{I}_{ca}^* = 240 \angle -240^\circ \times 15 \angle 210^\circ = 3117.69 - j1800 \text{ VA}$$

$$\text{So, } \vec{S}_{\text{total}} = \vec{S}_{ab} + \vec{S}_{bc} + \vec{S}_{ca}$$

$$= 7821.69 + j2356.92 \text{ VA} = 8169.08 \angle 16.77^\circ \text{ VA}$$

$$\text{So, } P_{\text{total}} = 7821.69 \text{ watt}$$

$$Q_{\text{total}} = 2356.92 \text{ VAR}$$

$$S_{\text{total}} = 8169.08 \text{ VA}$$

Measurement of 3-phase power:

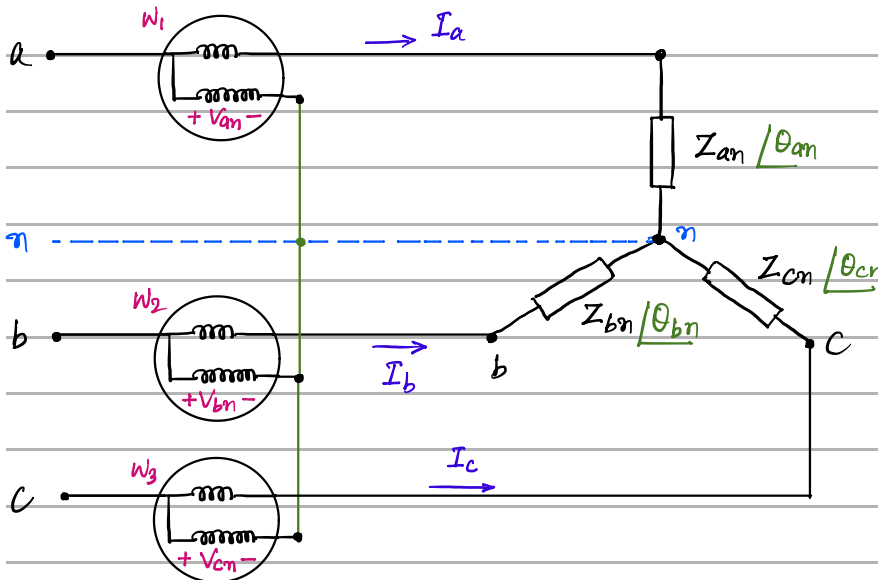
star ✓

delta ✗

$\left[\begin{array}{l} \text{star} \rightarrow Y \\ \text{delta} \rightarrow \Delta \end{array} \right]$

(i) Three-wattmeter method:

To measure power to a 4-wire (3 line wire + 1 neutral) star connected load require one wattmeter per phase. So, a total of 3 wattmeters are required in this method.



Reading of wattmeter W_1 ,

$$P_1 = V_{an} I_a \cos \theta_{an}$$

= power to phase an

Similarly,

$$P_2 = V_{bn} I_b \cos \theta_{bn}$$

$$P_3 = V_{cn} I_c \cos \theta_{cn}$$

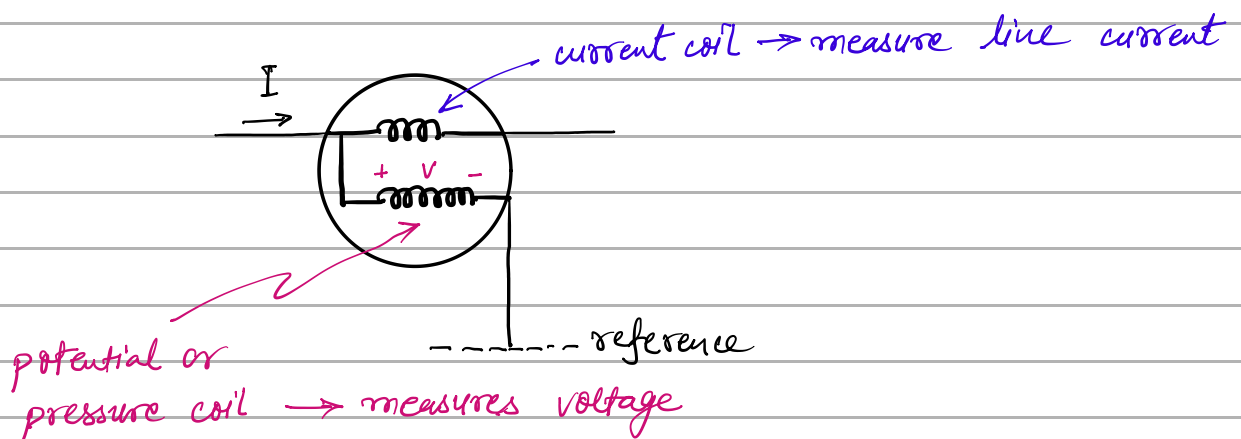
So, total power,

$$P_T = P_1 + P_2 + P_3$$

Note: If the circuit is 3-phase balanced circuit, then

$$P_T = 3P_1 = 3P_2 = 3P_3 \quad \text{i.e., } P_1 = P_2 = P_3$$

⇒ Only one wattmeter will be required.

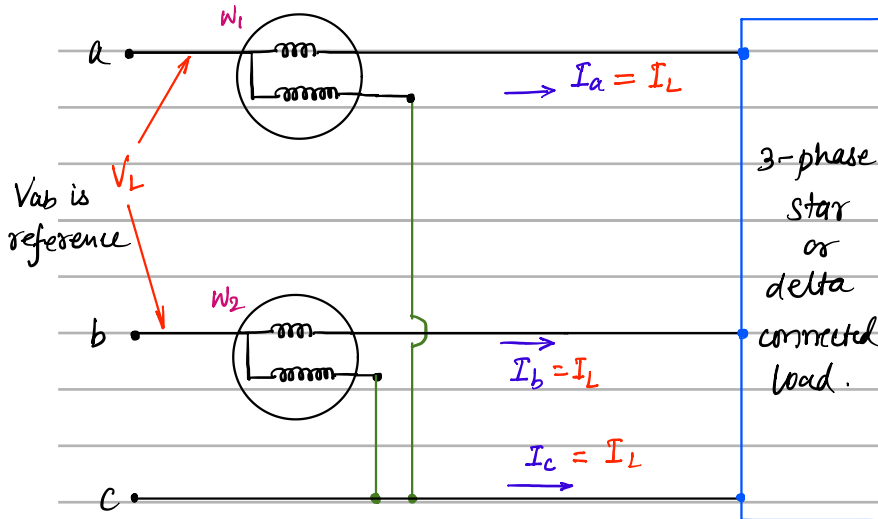


Imp

(ii) Two-wattmeter method:

star delta

To measure power for a three-wire (star & delta) system, only two wattmeters are required. Therefore load may be star or delta connected which may be balanced or unbalanced. The connection is shown below:



* The two wattmeters may be connected in any pair of lines with voltage terminals connected to the third line.

* The total real power is the algebraic sum of the meter readings.

Reading of wattmeter w_1 :

$$P_1 = V_L I_L \cos(30^\circ + \phi)$$

Reading of wattmeter w_2 :

$$P_2 = V_L I_L \cos(30^\circ - \phi)$$

Total real power consumed by 3-phase system (Δ or $\star$),

$$P = P_1 + P_2$$

$$= V_L I_L \cos(30^\circ + \phi)$$

$$+ V_L I_L \cos(30^\circ - \phi)$$

$$= \sqrt{3} V_L I_L \cos \phi$$

Power factor angle is given by,

$$\phi = \tan^{-1} \left[\frac{\sqrt{3}(P_2 - P_1)}{(P_1 + P_2)} \right] = \tan^{-1} \left(\frac{\text{reactive power}}{\text{active power}} \right)$$

So, operating power factor = $\cos \phi$

Total reactive power consumed, $Q = \sqrt{3} (P_1 - P_2)$

Q The readings of two wattmeters connected to measure the total power in a 3-phase, 400 V, 50 Hz system are 4 kW and 6 kW. Calculate the active power and reactive power consumed. Also find the operating power factor.

Solⁿ:

$$P_1 = 4000 \text{ W}$$

$$P_2 = 6000 \text{ W}$$

$$\begin{aligned}\therefore \text{Active or real power consumed by load, } P &= P_1 + P_2 \\ &= 4000 + 6000 \text{ W} \\ &= 10000 \text{ W} \\ &= 10 \text{ kW} \end{aligned}$$

$$\begin{aligned}\text{power factor angle, } \phi &= \tan^{-1} \left[\frac{\sqrt{3}(P_2 - P_1)}{P_1 + P_2} \right] \\ \Rightarrow \phi &= \tan^{-1} \left[\frac{\sqrt{3}(6000 - 4000)}{4000 + 6000} \right]\end{aligned}$$

$$= 19.106^\circ$$

$$\therefore \text{power factor} = \cos \phi = \cos (19.106^\circ) = 0.945$$

$$\begin{aligned}\text{And, reactive power, } Q &= S \sin \phi \\ &= \frac{P}{\cos \phi} \cdot \sin \phi = P \tan \phi \\ &= 10000 \times \tan (19.106^\circ) \\ &= 3463.98 \text{ VAR} \\ &= 3.46 \text{ KVAR} \end{aligned} \quad \left. \begin{array}{l} \text{OR,} \\ Q = \sqrt{3}(P_2 - P_1) \\ = \sqrt{3}(6000 - 4000) \\ = 2000\sqrt{3} \\ \approx 3.46 \text{ KVAR} \end{array} \right\}$$

Special conditions in 2-wattmeter method :

$$\phi = \tan^{-1} \frac{\sqrt{3}(P_2 - P_1)}{P_1 + P_2} \quad \text{and, } Q = \sqrt{3}(P_2 - P_1)$$

(i) Both meter reading are equal :

$$\text{i.e., } P_1 = P_2$$

$$\therefore \phi = \tan^{-1}(0) = 0 \Rightarrow \cos \phi = 1$$

Conclusion : Load is operating at unity power factor. Each wattmeter reads half the total power.

(ii) Wattmeter W_1 reads greater than wattmeter W_2 :

$$\text{i.e., } P_1 > P_2$$

$$\therefore \phi = \tan^{-1}(-ve) = -ve \text{ angle} \Rightarrow \text{leading pf}$$

$$\text{and, } Q = \sqrt{3}(P_2 - P_1) = -ve \Rightarrow \text{capacitive circuit} \\ \Rightarrow \text{leading pf}$$

Conclusion : Load is operating at leading pf $\Rightarrow$ capacitive load.

(iii) Wattmeter W_2 reads greater than wattmeter W_1 :

Conclusion : Load is operating at lagging pf $\Rightarrow$ inductive load.

(iv) If the load is purely capacitive or inductive :

$$\phi = \tan^{-1} \left[\frac{\sqrt{3}(P_1 - P_2)}{P_1 + P_2} \right] = 90^\circ$$

$$\Rightarrow P_1 = -P_2$$

Conclusion : At reading of two wattmeters are equal but of opposite sign.

(V) when one of the two wattmeters reads zero :

i.e., if $P_1 = 0$,

$$\phi = \tan^{-1} \left(\frac{\sqrt{3}(P_2 - P_1)}{P_1 + P_2} \right) = \tan^{-1} (\sqrt{3}) = 60^\circ$$

$$\Rightarrow \cos \phi = \cos 60^\circ = 0.5$$

Conclusion : The load operates at 0.5 pf.

Q Two wattmeters are connected to measure the power consumed by a 3-phase balanced load. One of the wattmeters read 1500 W and the other reads 700 W. Find the power factor of the load, when (a) both readings are positive, and (b) when the reading of second wattmeter is obtained after reversing its current coil connection.

Soln :

(a) $P_1 = 1500 \text{ W}$

$$P_2 = 700 \text{ W}$$

$$\therefore \phi = \tan^{-1} \left[\frac{\sqrt{3} (P_1 - P_2)}{P_1 + P_2} \right] = \tan^{-1} \left[\frac{\sqrt{3} (1500 - 700)}{1500 + 700} \right]$$

$$= 32.2^\circ$$

$$\therefore \text{pf} = \cos \phi = \cos 32.2^\circ = 0.846$$

(b) $P_1 = 1500 \text{ W}$

$$P_2 = -700 \text{ W}$$

$$\therefore \phi = \tan^{-1} \left[\frac{\sqrt{3} (1500 - (-700))}{1500 + (-700)} \right] = \tan^{-1} \left(\frac{\sqrt{3} \times 2200}{800} \right)$$

$$= 78.1^\circ$$

$$\therefore \text{pf} = \cos \phi = \cos 78.1^\circ = 0.21$$

Q Two wattmeters are connected to measure the power consumed by a 3-phase load with power factor 0.4. Total power consumed by the load, as indicated by the two wattmeters is 30 kW. Find the individual wattmeter readings.

Solⁿ:

Clearly, $P_1 + P_2 = 30000$ — (i)

And, $\phi = \cos^{-1}(0.4) = \tan^{-1} \left[\frac{\sqrt{3} (P_1 - P_2)}{P_1 + P_2} \right]$

$$\Rightarrow \frac{\sqrt{3} (P_1 - P_2)}{P_1 + P_2} = \tan [\cos^{-1}(0.4)]$$

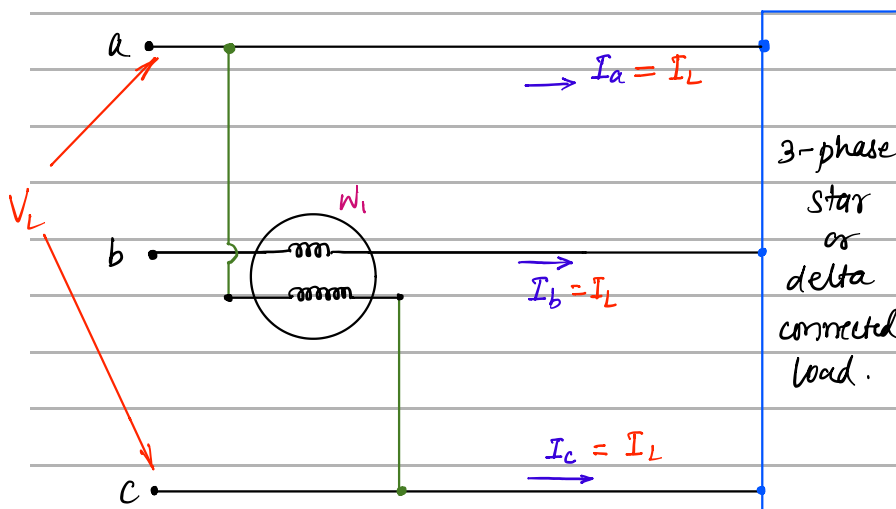
$$\Rightarrow P_1 - P_2 = \frac{30000}{\sqrt{3}} \tan [\cos^{-1}(0.4)]$$

$$\Rightarrow P_1 - P_2 = 39686.27$$
 — (ii)

Solving (i) and (ii), we get

$$\left. \begin{aligned} P_1 &= 34843.135 \text{ W} = 34.84 \text{ kW} \\ P_2 &= -4843.135 \text{ W} = -4.84 \text{ kW} \end{aligned} \right\} \text{ Ans}$$

Measurement of reactive power using single wattmeter:



Reading of wattmeter

$$= V_{ac} I_b \cos(90^\circ + \phi)$$

$$= V_L I_L \cos(90^\circ + \phi)$$

$$= V_L I_L \sin \phi$$

So, the required reactive power is calculated as

$$\sqrt{3} V_L I_L \sin \phi$$